

Final Assignment: Data Engineering for a Consulting Firm



Estimated time needed: 45 mins

You are a data engineer at a data analytics consulting company. Your company prides itself in being able to efficiently handle data in any format on any database on any platform. Analysts in your office need to work with data on different databases, and data in different formats. While these analysts are good at analyzing data, they count on you to be able to move data from external sources into various databases, to be able to move data from one type of database to another, and be able to run basic queries on various databases.

Objectives

In this assignment you will:

- Import data into a MongoDB database.
- Query data in a MongoDB database.
- Export data from MongoDB.
- Import data into a Cassandra database.
- Query data in a Cassandra database.

Optional: After completing each task, save both the executed commands and their corresponding outputs in a text document, or take screenshots—depending on the option you have chosen. This information will be required to answer questions during the final project submission.

About Skills Network Cloud IDE

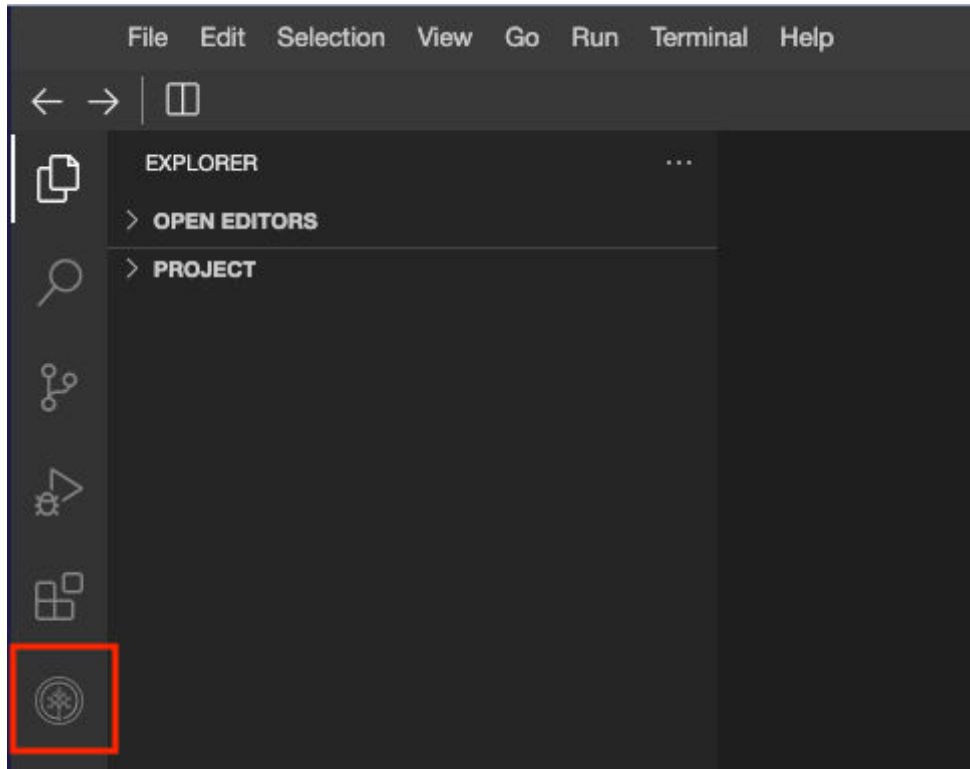
Skills Network Cloud IDE (based on Theia and Docker) provides an environment for hands on labs for course and project related labs. Theia is an open source IDE (Integrated Development Environment), that can be run on desktop or on the cloud. to complete this lab, we will be using the Cloud IDE based on Theia and MongoDB/Cassandra running in a Docker container.

Important Notice about this lab environment

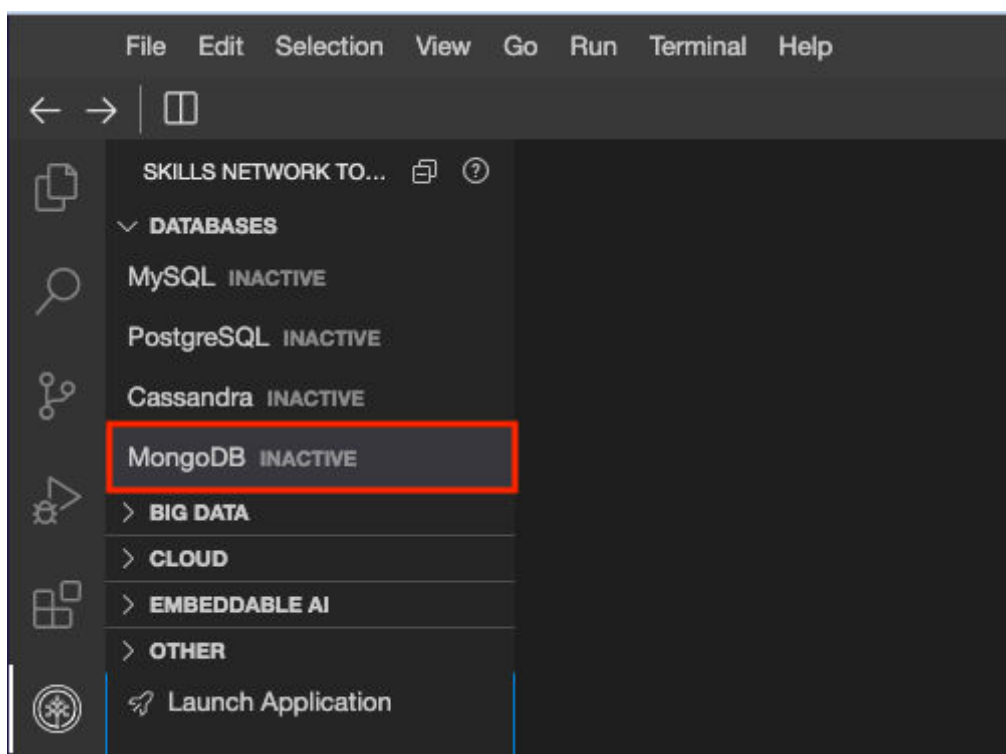
Please be aware that sessions for this lab environment are not persisted. Every time you connect to this lab, a new environment is created for you. Any data you may have saved in the earlier session would get lost. Plan to complete these labs in a single session, to avoid losing your data.

Set-up: Start MongoDB

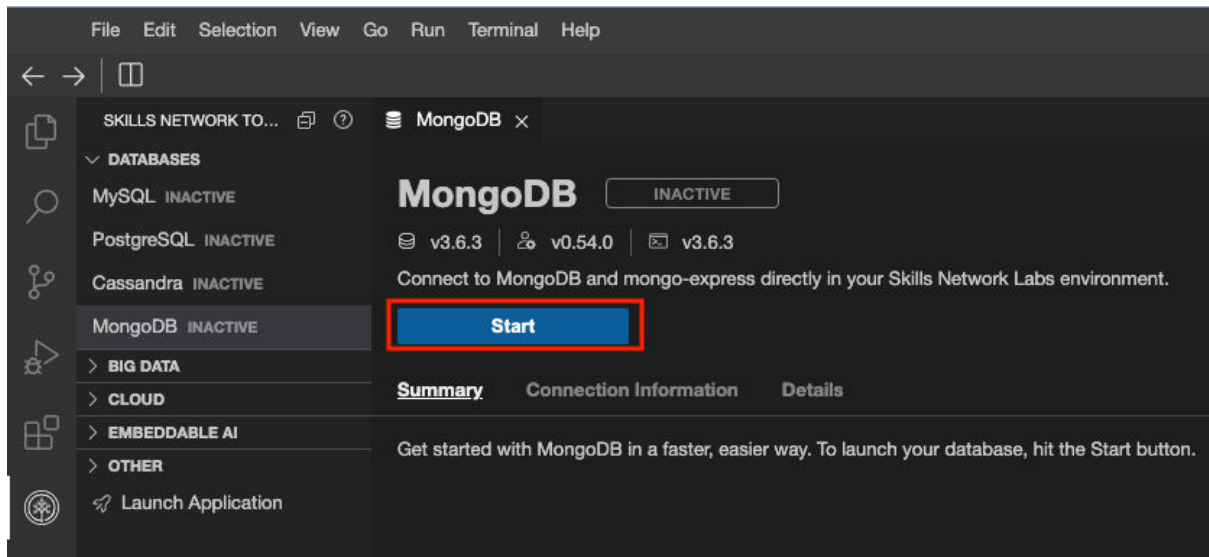
Navigate to Skills Network Toolbox.



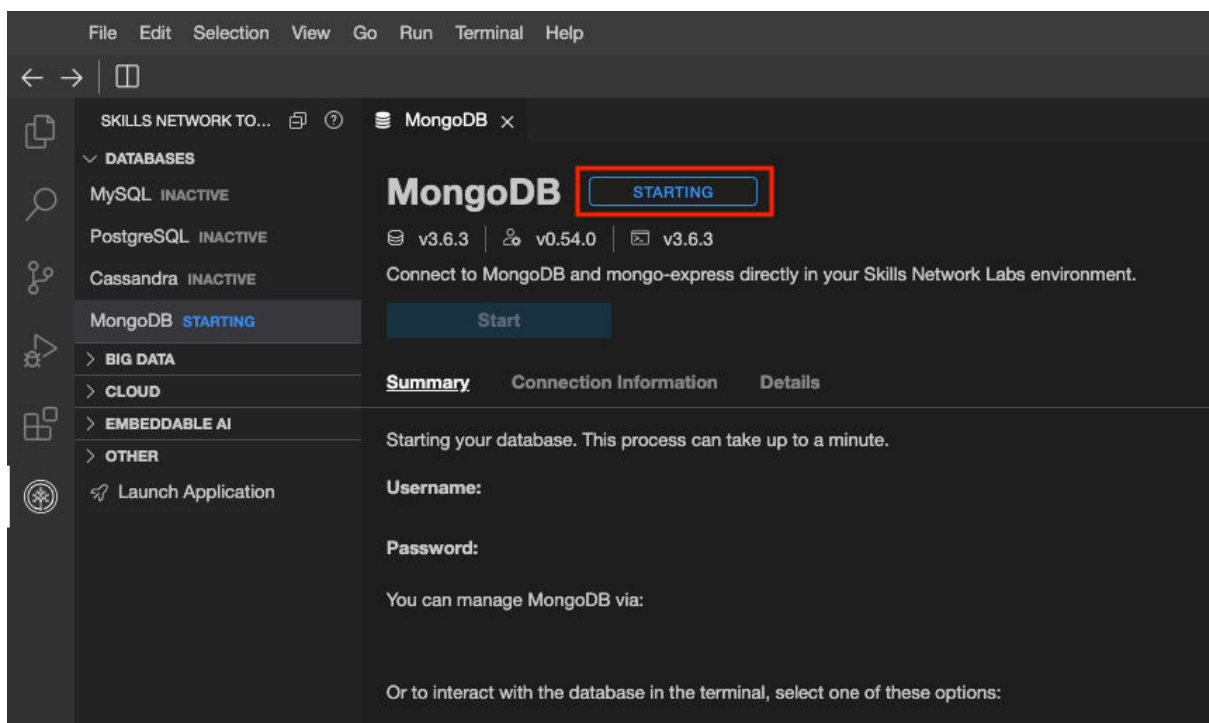
You will notice MongoDB listed there, but inactive, which means the database is not available to use.



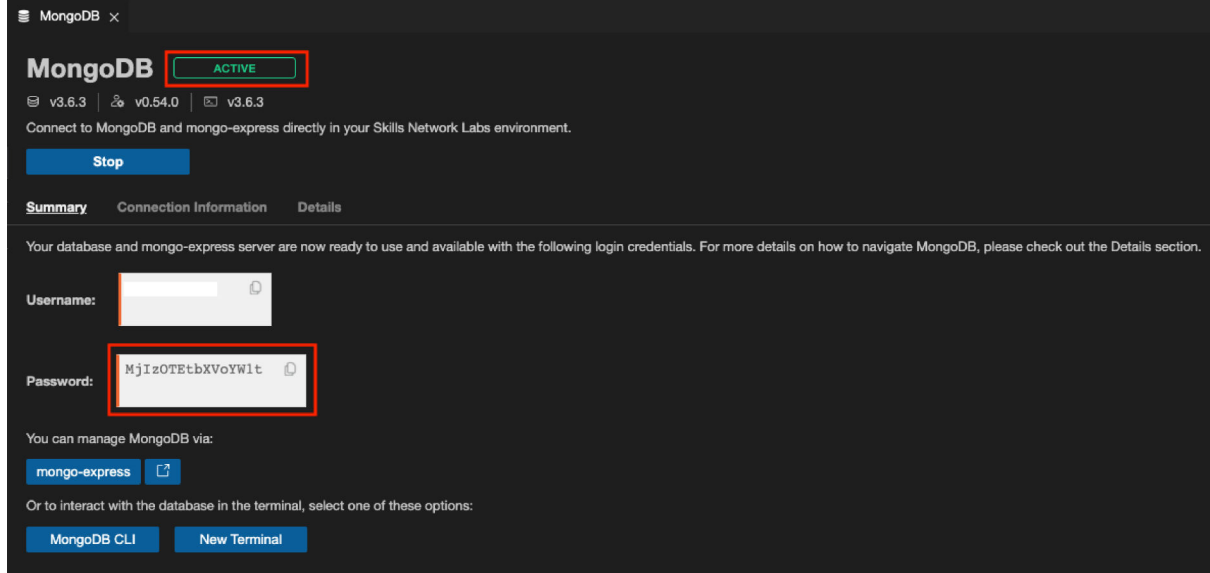
Once you click on the database, you will see more details and a button to start the database.



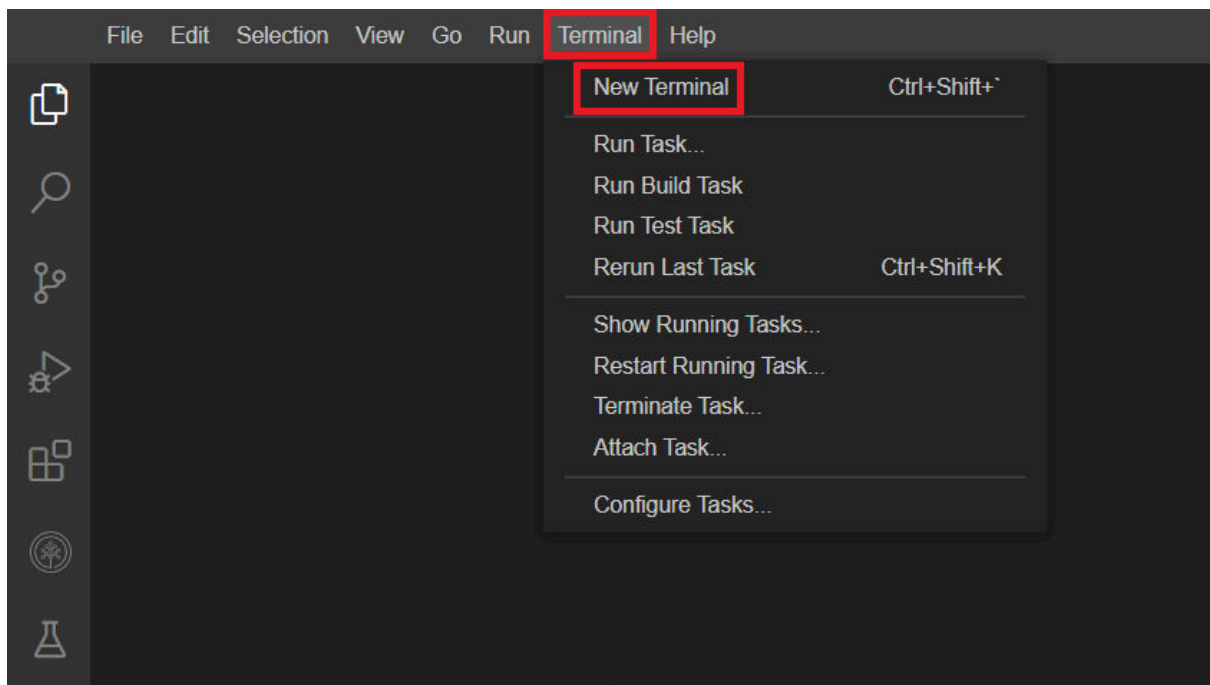
Clicking the start button runs a background process to configure and start your MongoDB server.



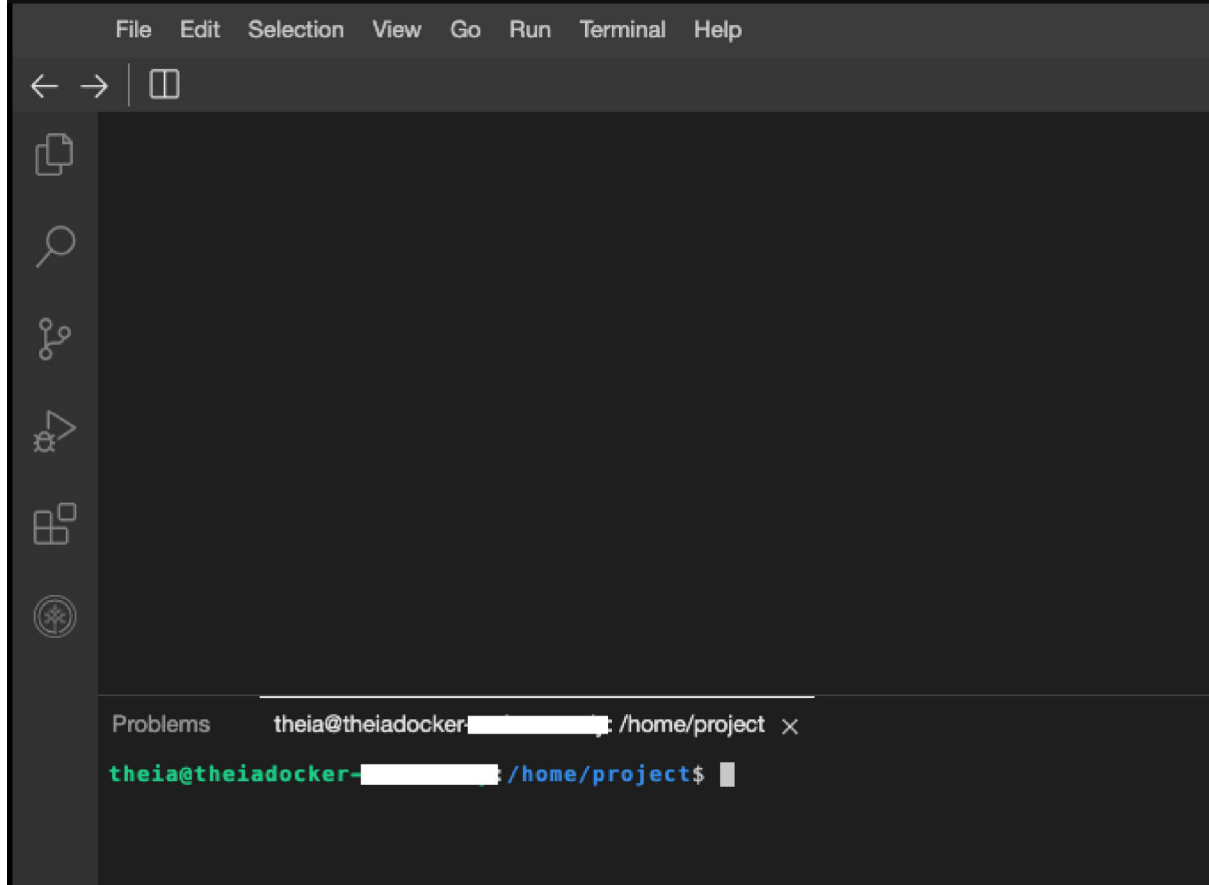
Soon, your server is ready for use. This deployment has access control enabled and MongoDB enforces authentication. So, take note of the password. You will need this password to login as **root** user.



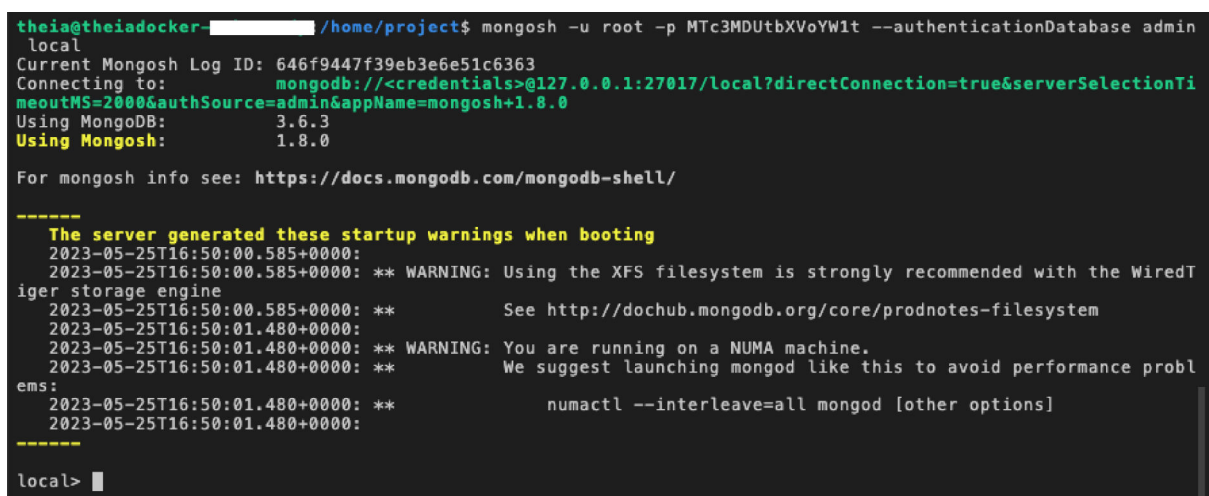
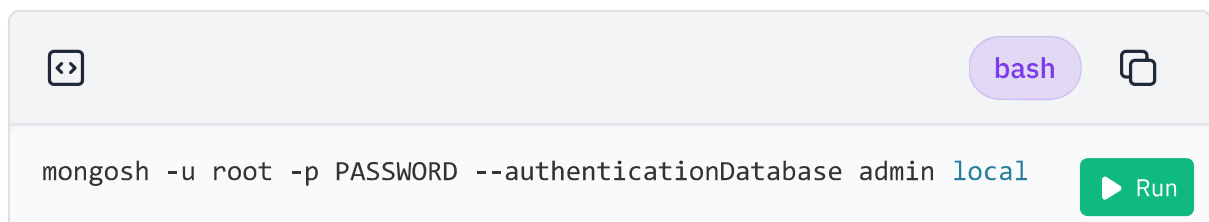
You can now open a new terminal window.



The new terminal window opens at the bottom of the screen as seen in the following image.

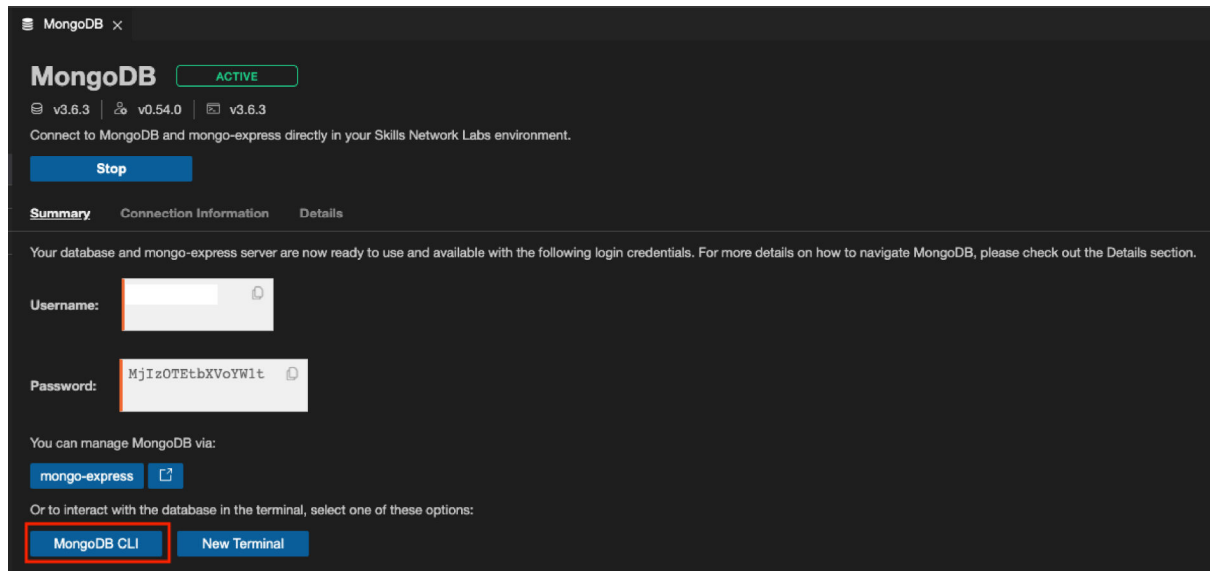


Run the following command in the newly opened terminal window. (You can copy the code by clicking on the copy button on the bottom right of the following codeblock and then paste the code where needed.)

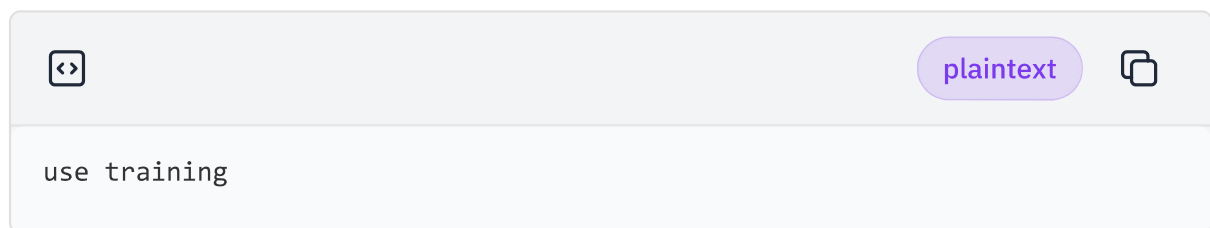


The command contains the username and password to connect to mongodb server (the text after the -p option is the password). Your output would be different from the output shown in the prior image. Copy the command given to you, and keep this

Or, you can simply click on MongoDB CLI



In MongoDB CLI (the mongo shell), switch the context to the **training** database.

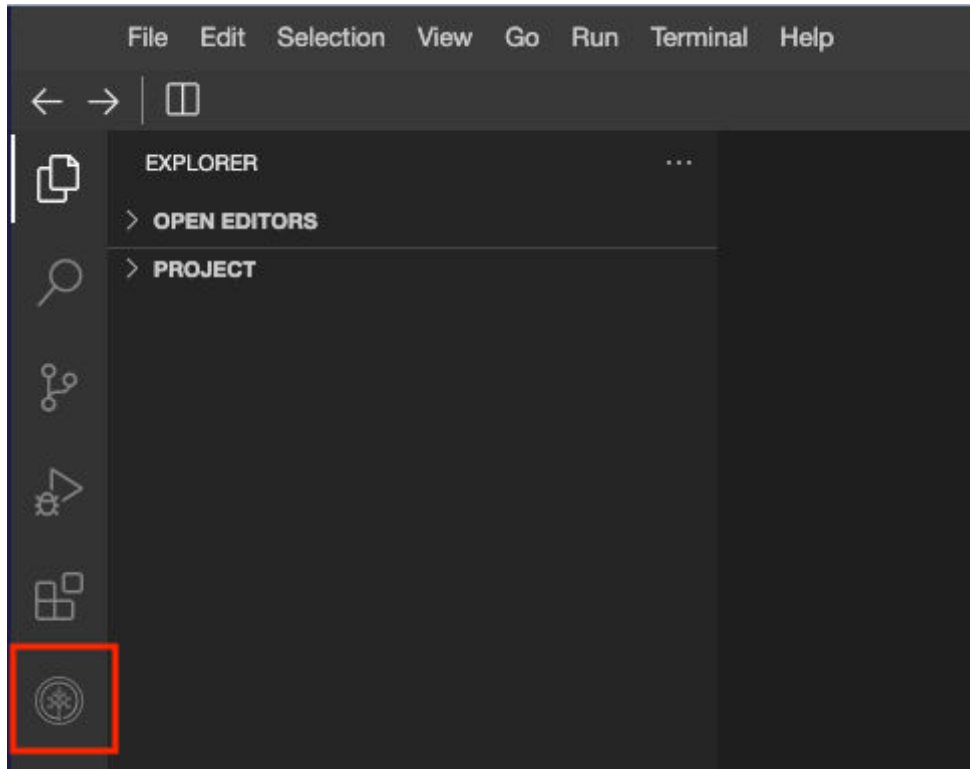


Create a collection called **bigdata**

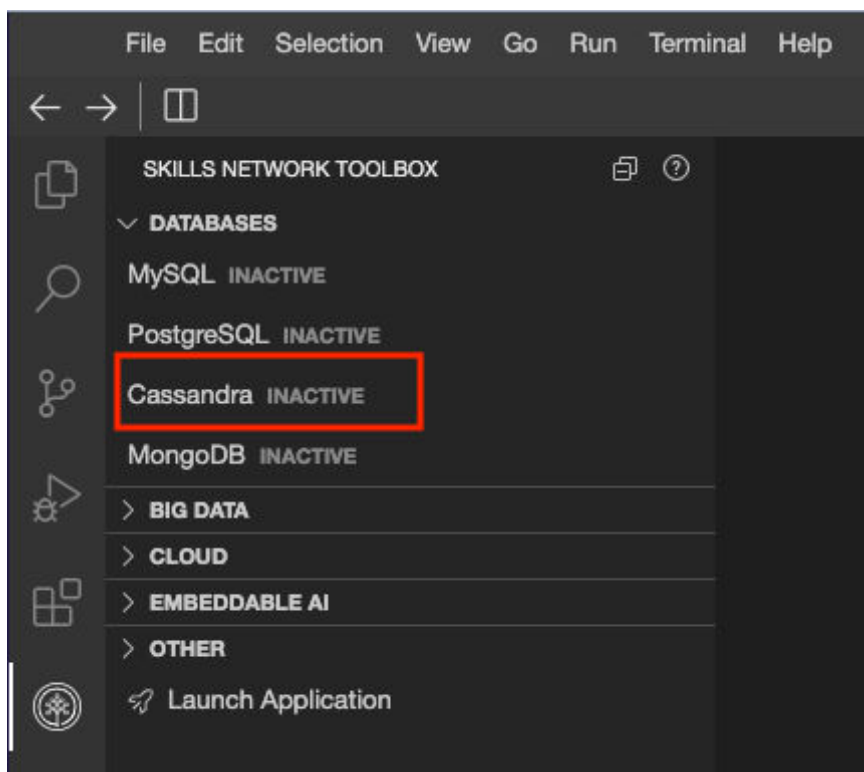


Set-up: Start Cassandra

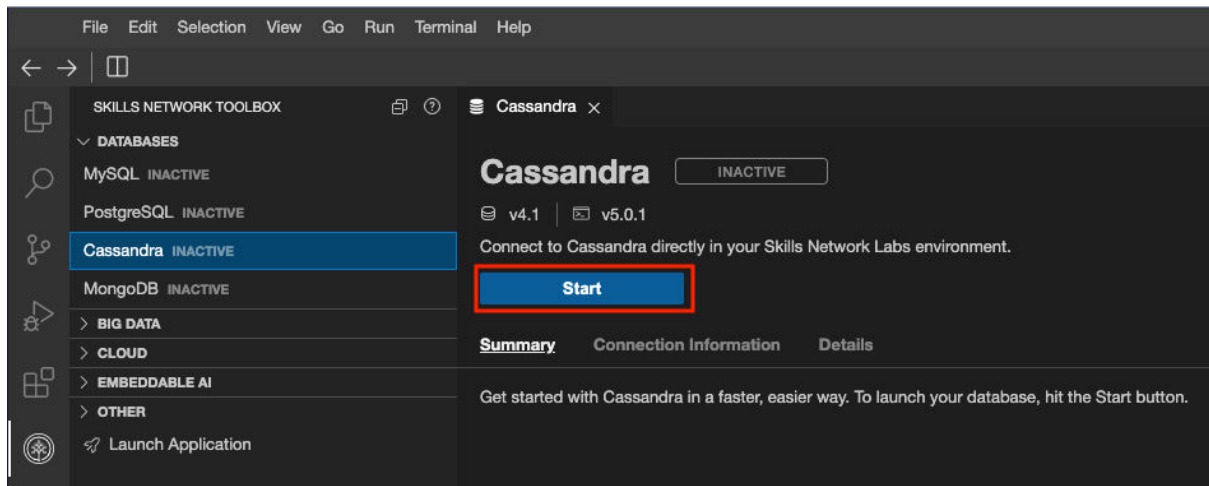
Navigate to Skills Network Toolbox.



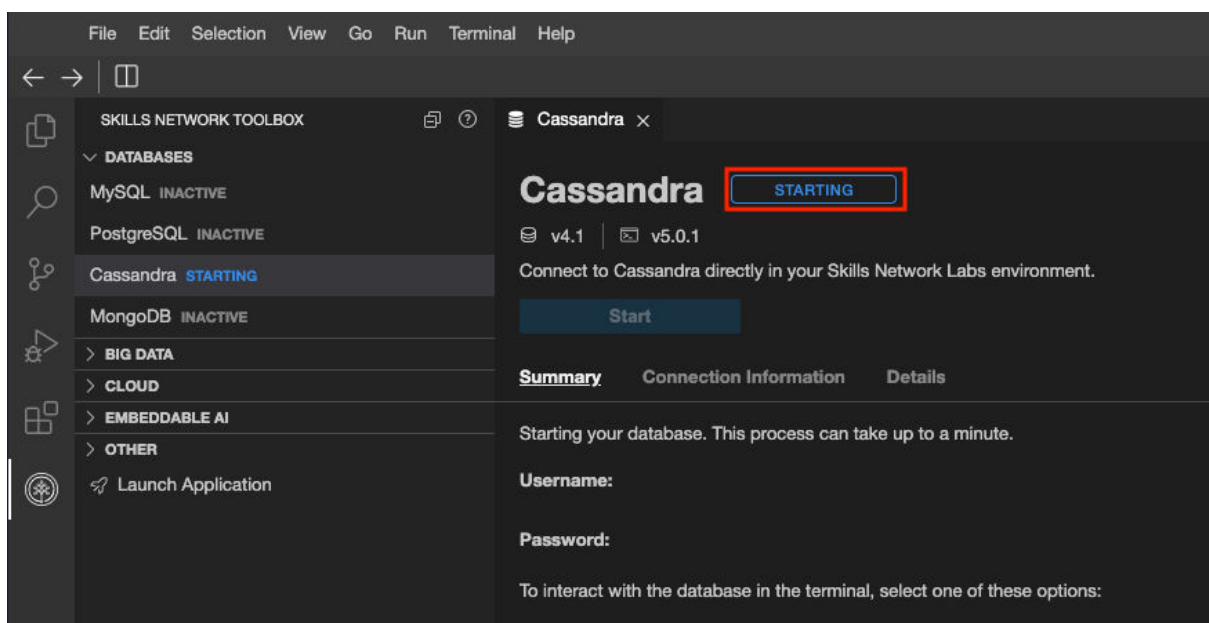
Cassandra is listed, but inactive, which means a database is not available to use.



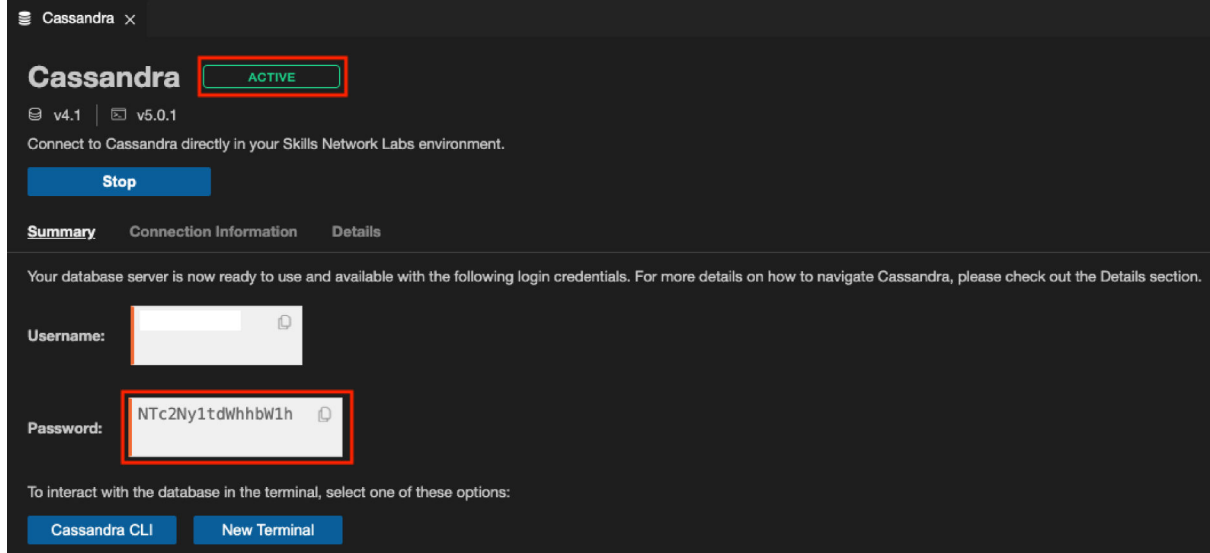
Once you click on the database, you will see more details and the button to start the database.



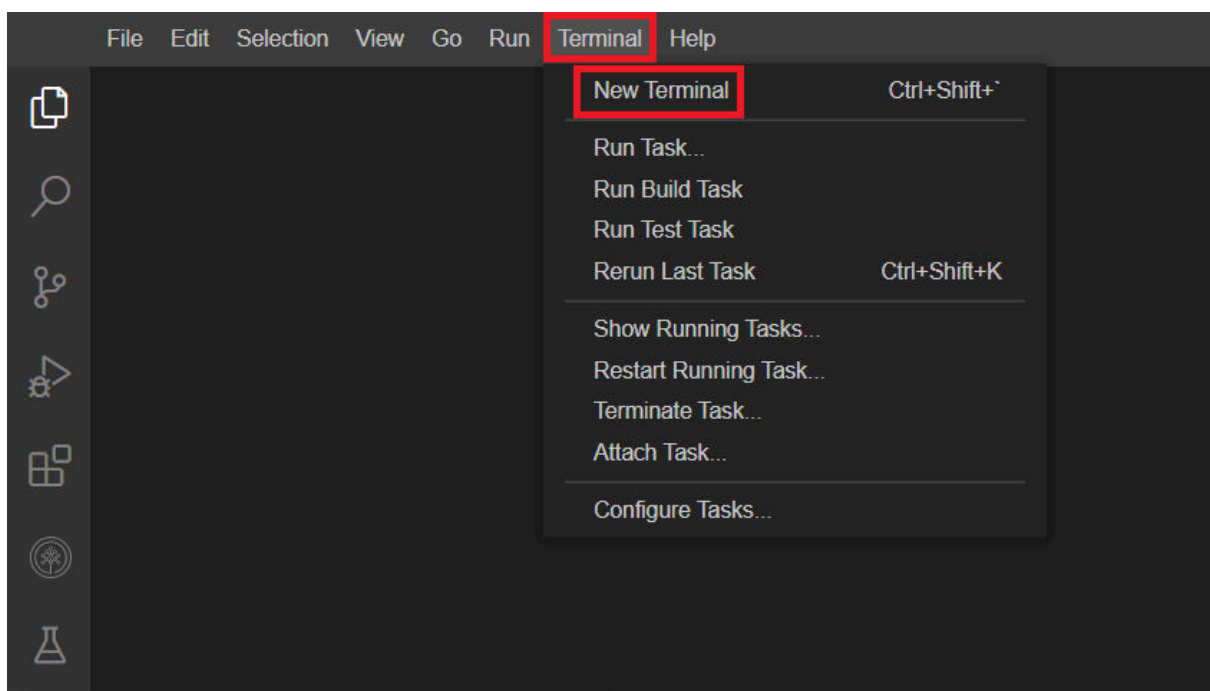
Clicking the Start button runs a background process to configure and start your Cassandra server.



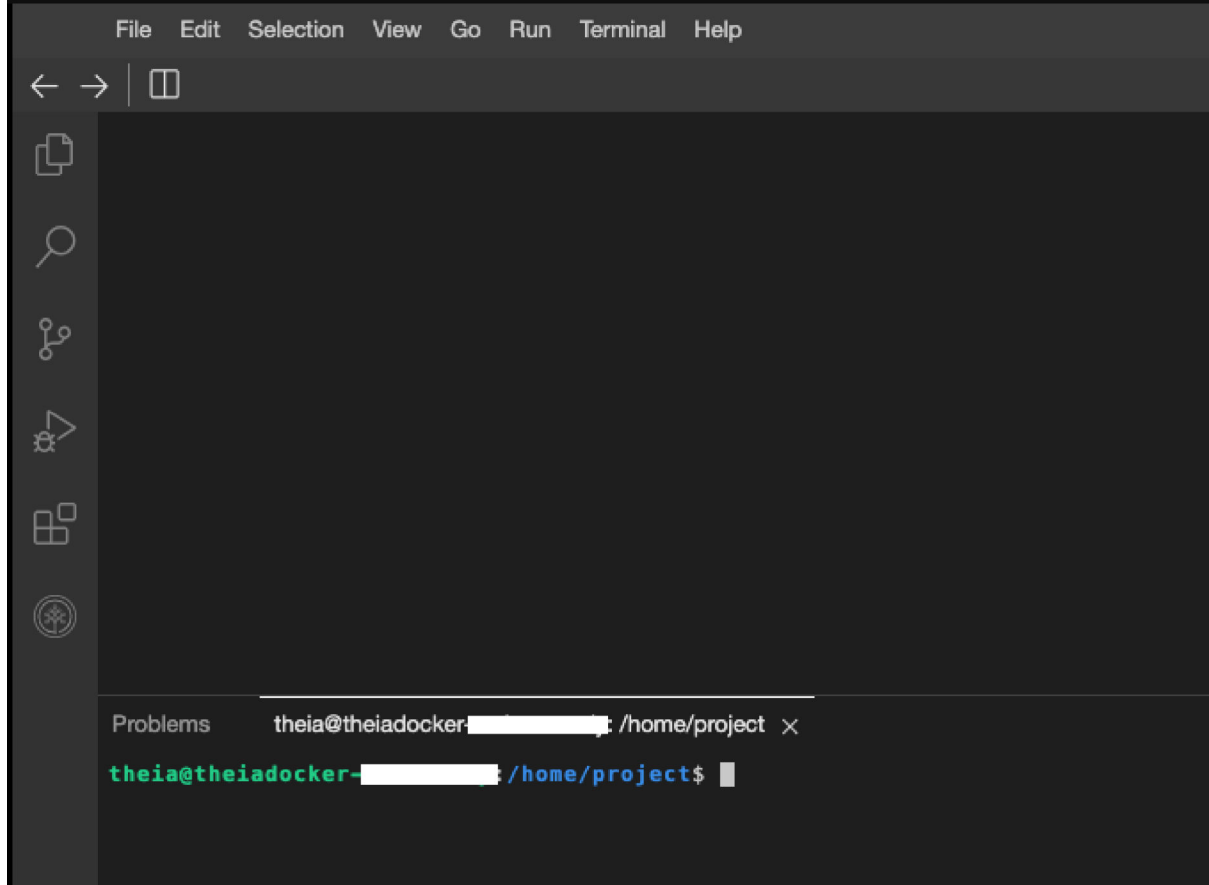
Shortly after that, your server is ready for use. This deployment has access control enabled and Cassandra enforces authentication. So, take note of the password as you will need the password to login as a `cassandra` user.



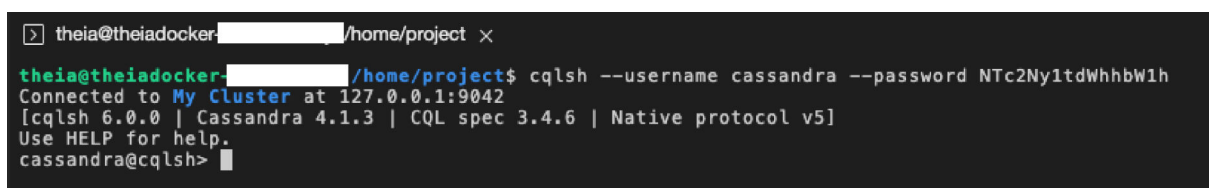
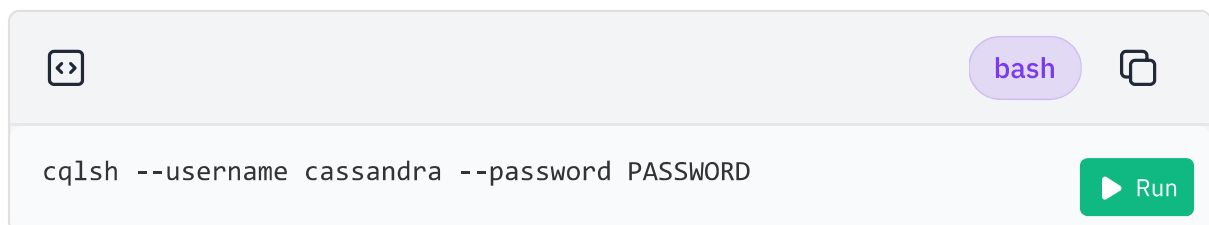
You can now type `\ open terminal` and enter the details.



This action open a new terminal at the bottom of the screen as seen in the following image.



Run the following command in the newly opened terminal. (You can copy the code by clicking on the little copy button on the bottom right of the codeblock below and then paste the code where needed.)



This command contains the username and password to connect to Cassandra server. Your output could be different from the one shown above. Copy the command given to you, and keep the command handy. You will use this command in the next step.

Or you can simply click on Cassandra CLI which does that for you.

Cassandra

ACTIVE

v4.1 | v5.0.1

Connect to Cassandra directly in your Skills Network Labs environment.

Stop

Summary

Connection Information

Details

Your database server is now ready to use and available with the following login credentials:

Username:

Password:

NTc2Ny1tdWhhbW1h

To interact with the database in the terminal, select one of these options:

Cassandra CLI

New Terminal

Exercise 1: Working with a MongoDB database

Download sample data file

Use the following command to download the data file in your Cloud IDE project directory.

```
 bash   
curl -O https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud
```

A sample movie document

```
 json   
{  
  _id: '9',  
  title: 'The Lost City of Z',  
  genre: 'Action,Adventure,Biography',  
  Description: 'A true-life drama, centering on British explorer Col. Perc  
  Director: 'James Gray',  
  Actors: 'Charlie Hunnam, Robert Pattinson, Sienna Miller, Tom Holland',  
  year: 2016,  
  'Runtime (Minutes)': 141,  
  rating: 'unrated',  
  Votes: 7188,  
  'Revenue (Millions)': 8.01,  
  Metascore: 78  
}
```

Task 1: Import `movies.json` into mongodb server into a database named `entertainment` and a collection named `movies`.

Now import the data from the `movies.json` file into MongoDB and verify that it is

Take a screenshot of the command you used and its output.

Task 2: Write a mongodb query to find the year in which most number of movies were released

Write and execute an aggregation query to count movies by release year, sort the results in descending order, and return the year with the highest count.

► [Click here for Hint](#)

Task 3: Write a mongodb query to find the count of movies released after the year 1999

Write and execute the query and verify that the output shows 99 movies released after 1999.

Task 4: Write a query to find out the average votes for movies released in 2007

Write and execute the query and verify that the output shows an average vote count of 192.5 for movies released in 2007.

► [Click here for Hint](#)

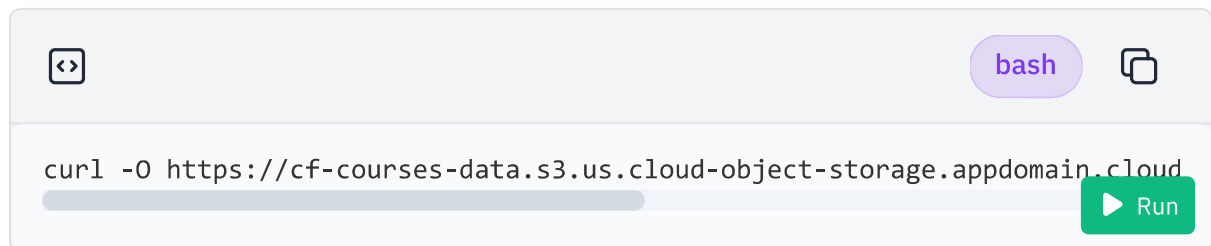
Task 5: Export the fields `_id`, `title`, `year`, `rating` and `Director` from the `movies` collection into a file named `partial_data.csv`

Take a screenshot of the command you used and its output.

Exercise 2 - Working with a Cassandra database

Download sample data file

If you haven't managed to successfully export data into `partial_data.csv` file, then download the data file already created for you with the following command.



```
curl -O https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud
```

The image shows a terminal window with a light gray header bar containing a code icon, a 'bash' label, and a copy icon. The terminal area has a light blue background and displays the command `curl -O https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud`. A green 'Run' button is located at the bottom right of the terminal area.

Task 6 - Create a keyspace named `entertainment`

Take a screenshot of the command you used and its output.

▶ Click here for Hint

Once created, use the `describe` command and use the output for assessment.

Task 7 - Import `partial_data.csv` into cassandra server into a keyspace named `entertainment` and a table named `movies`

While creating the table `movies` configure all of the columns as text columns including the id column.

- `_id`
- `title`
- `year`
- `rating`
- `director`

▶ [Click here for Hint](#)

Take a screenshot of the command you used and its output.

And now import the data present in `partial_data.csv`

▶ [Click here for Hint](#)

Task 8 - Write a cql query to count the number of rows in the `movies` table

Write and execute the query and verify that the output displays the total number of rows in the movies table.

▶ [Click here for Hint](#)

Task 9 - Create an index for the `rating` column in the `movies` table using cql

Take a screenshot of the command you used and its output.

▶ [Click here for Hint](#)

And then run the `describe` on `movies` that shows `CREATE INDEX` listed in the output.

Task 10 - Write a cql query to count the number of movies that are rated 'G'.

Write and execute the query and verify that the output displays the total number of movies with a 'G' rating.

▶ [Click here for Hint](#)

Congratulations! That's a wrap!

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